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TEMPLEs: Teacher Evaluation using Metamorphic-mediated Personality-aware LLM Evaluators

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***Abstract*—This paper proposes a novel framework for evaluating teaching quality in higher education (HE) through the use of multiple large language models (LLMs). The paper draws an analogy between the distinct "personalities" of different LLMs and the varied perspectives of human peer reviewers. This analogy is used to explore the inherent difficulty in objectively assessing teaching effectiveness, which we describe as an instance of the "oracle problem" — a well-studied area of software testing/engineering where categorical determination of correctness may not be possible. Traditional teaching/teacher evaluation methods are notoriously subjective and biased. Our proposed solution uses a suite of LLMs as a "panel of peer reviewers," taking advantage of their diverse analytical styles (personalities). The framework incorporates a validation layer based on metamorphic testing (MT), a software-testing approach with an excellent record of alleviating the oracle problem. Instead of focusing on individual correctness, MT uses relations (called metamorphic relations, MRs) to help identify errors or mistakes. This work is contextualised within the recent, rapid adoption of generative AI in HE, particularly within Sino-foreign HE institutions (SfHEIs). A key aim of this project is to package this framework into a dual-purpose open educational resource (OER): a formal tool to support teaching/teacher evaluation, and a resource to help educators, both in-service and pre-service, with their self-reflection and continuous professional development (CPD).**

***Index Terms*—Teacher/teaching evaluation, higher education (HE), Sino-foreign HE, metamorphic testing/thinking, open educational resources (OERs), continuous professional development (CPD)**

I. Introduction

The evaluation of teaching quality, and hence teachers, in higher education (HE) represents a persistent and formidable challenge, a Gordian knot [1] at the centre of academic administration and professional development. The stakes are profoundly high, influencing not only the career trajectories of educators, but also institutional reputation and, most importantly, the quality of the student educational experience. At its core, the difficulty lies in the absence of a definitive, objective standard — a "ground truth" — for what constitutes effective teaching: This is a major challenge to ensuring and enhancing teacher and teaching quality [2].

In the field of software engineering (SE), a test oracle is a mechanism, human or automated, that can definitively determine whether or not a system has behaved correctly for a given input [3], [4]. For complex, non-deterministic systems — such as the majority of modern systems based on artificial intelligence (AI) and machine learning (ML) — a perfect oracle often does not exist: This situation, where there is no practicably applicable oracle, has been called the oracle problem [4]. Software systems facing the oracle problem have been called untestable [5], [6]. Similar to software systems, a system for the evaluation and assessment of teaching may also be considered to face the oracle problem. A teacher's performance is not a simple, deterministic function that produces a single, verifiable output. Teaching effectiveness (both actual and perceived) is a complex interplay of variables: It is highly contextual, deeply multifaceted, and inherently subjective [7]–[9]. The quality of a seminar on post-structuralist theory [10], for example, cannot be measured in the same way as a laboratory session on organic chemistry [11]. Likewise, the needs of first-year undergraduates are vastly different from those of doctoral candidates [12]. Finally, pedagogical approaches that may thrive in a small, discussion-based liberal arts college may falter in a large, research-intensive institution [13]–[15].

By reframing teaching evaluation as an oracle problem, we aim to apply insights from SE to the issue [16]. Traditionally, metamorphic testing (MT) [17], [18], and its related concepts of metamorphic exploration [19] and thinking [20], have been shown to very effectively alleviate the oracle problem, and enable the testing of untestable systems [6], [21]. For too long, HE has relied on two primary modalities — peer observation of teaching (PoT) [22]–[24] and student evaluations of teaching/teachers (SETs) [2], [25], [26]: Both of these face systemic issues of subjectivity, bias, and unreliability, making them fundamentally flawed tools [2], [27]. There is an urgent need for an innovative approach, one that can provide more robust, reliable, and fair assessments [28]. This paper

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argues for a paradigm shift: A move away from the futile search for a single, authoritative judge, and towards a new, multi-perspectival evaluation model [29], [30]. We propose leveraging the power of generative AI not as a single, supposedly objective new oracle, but as a diverse and deliberately assembled panel of synthetic reviewers. The panel's outputs (assessments and judgements) can be examined through the lens of MT, providing a new perspective on teacher and teaching evaluation.

II. Review of Traditional Evaluation Modalities

Scientific rigour tells us that relying on any single source of evidence is inadequate [38]. Table I summarises and compares PoT and SETs with a proposed LLM-based teaching evaluation approach. The systemic failures of both PoT and SETs highlight the need for a new paradigm.

A. Peer Observation of Teaching (PoT)

PoT, where colleagues observe each other's classes, is a widely accepted tool and approach [22]–[24]. However, its implementation faces challenges that undermine its validity: Subjectivity and, critically, the lack of adequate training for observers, remain key issues [39]: This often results in a practice shrouded in uncertainty. A core weakness is the potential for bias [33]: Instructors often select their own observers, introducing influences from personal relationships, or a fear of negative interactions [40].

The frequent lack of formal training for PoT compounds the bias difficulties. Unlike in K-12 education, HE teachers often have little to no expertise in systematic observation, leading to highly variable interpretations of evaluation rubrics, and ultimate assessments [41]. PoT also often faces a tension between its dual purposes: formative development (improvement) and summative evaluation (for promotion and tenure, amongst others). This conflict can neutralise its effectiveness, with feedback often becoming confirmatory rather than genuinely transformative [40].

B. Student Evaluation of Teaching/Teachers (SETs)

SETs are perhaps the most widely used form of teaching evaluation [2], [25], [26]. However, a growing body of evidence highlights how deeply flawed a metric they can be [42]. A core issue is their weak (or non-existent) correlation with objective measures of student learning. SETs have been reported to be influenced by variables unrelated to instruction quality, such as class size, discipline, and students' anticipated grades. This can create a perverse incentive for lenient grading to (hopefully) receive positive SETs [43].

A very troubling aspect of SETs is the well-documented issue of equity bias [32], [42], [44]. Numerous studies have shown that women and faculty of colour receive significantly lower scores than their white male colleagues, even when all teaching behaviours are held constant [45]. This gender bias can account for as much as a half-point difference on a five-point scale [46] — a margin that can be career-altering.

There are many aspects of a student's educational experience that students themselves are not fully equipped or qualified to assess [47]: Core components of pedagogical design, and teacher effectiveness, arguably, are among the most obvious. The anonymity of SETs can be weaponised by disgruntled students, ... and even if not intentionally harmful, poorly-informed comments can have very negative impacts.

III. An LLM-based Multi-Perspectival Approach

The systemic failures of PoT and SETs require a new approach. The American Sociological Association, among others, argues that student ratings should never be used in isolation, but must be part of a broader assessment incorporating multiple forms of evidence [48]. This is reinforced by research on multisource assessment (360-degree feedback [49]), according to which, a diversity of perspectives [50], even contradictory ones, provides a more accurate picture than a single viewpoint. The equitable evaluation of teacher and teaching requires a framework that can systematically gather, weigh, and synthesise evidence from a diverse assembly of viewpoints. The recent advances in generative AI, including large language models (LLMs) like ChatGPT [51], suggest a possible solution direction.

A. Generative AI and LLMs

Generative AI is a category of AI that is capable of producing novel content, such as text, images, or computer code [52]. LLMs are a key type of generative AI, trained on very large datasets of text, which allows them to comprehend and generate language in a human-like manner [53]. Popular examples of current LLMs include ChatGPT[1], Claude[2], and Gemini[3].

An important characteristic of LLMs is their ability to exhibit distinct "personalities" [54]: Different models, or the same model guided by different instructions (known as prompts), can respond to identical queries with varied styles and perspectives, much like a panel of human experts would [55], [56]. For example, one LLM persona might adopt a highly critical stance, whereas another could provide more encouraging, developmental feedback.

This highlights a challenge analogous to the oracle problem in software testing. When using LLMs for evaluation, there is no definitive "oracle" to verify the correctness of their output. The quality of teaching is multifaceted and subjective: This means that there is no single, perfect assessment. Therefore, determining the accuracy of an LLM's evaluation is not straightforward.

Our proposed framework addresses this challenge by applying principles from MT. Rather than relying on a single LLM, we use a panel of LLMs with diverse personalities. We then define metamorphic relations (MRs)—properties that are expected to hold between the outputs of these different LLMs when the input (e.g., teaching materials) is changed in

[1]ChatGPT by OpenAI: https://chatgpt.com
[2]Claude by Anthropic: https://claude.ai
[3]Gemini by Google: https://gemini.google.com

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TABLE I
COMPARATIVE ANALYSIS OF TEACHING EVALUATION APPROACHES

Feature / Dimension	Student Evaluations of Teaching (SETs)	Peer Observation of Teaching (PoT)	TEMPLEs LLM-Panel Framework
Objectivity / Bias	Susceptible to biases related to instructor gender, race, and students' grade expectations [31]–[33].	Effectiveness is highly dependent on observer training, institutional context, and the quality of the observation protocol [34].	Aims to reduce human biases using a panel of synthetic reviewers/agents and an MT verification layer.
Scalability	High. Can be administered to large student populations with minimal marginal cost.	Low. Requires a significant investment of faculty time for both observation and providing feedback.	Very High. New agents/personalities can be introduced based on institutional context and teacher needs.
Reliability	Reliable estimates require aggregation across many courses [25].	Variable. Can be improved with clear criteria and rigorous training (calibration) for observers, but this is not always implemented [35].	High reliability is derived from an internal, MT-based verification loop that ensures the final output is robust, consistent, and logically sound.
Developmental Value	Generally considered low for formative improvement. The feedback is often not specific enough to be actionable [36].	Potentially high, especially in a formative context where it is decoupled from summative performance reviews [37].	Potentially High, offering tailored formative feedback for self-reflection and continuous professional development.

a specific way. For instance, an MR might state that if a lesson plan is modified to include more interactive elements, the evaluation scores from a majority of the LLM personas should increase. By verifying these relations, we can build confidence in the evaluation process without requiring an unobtainable oracle.

IV. THE TEMPLES FRAMEWORK

This section presents the vision for how a collection of LLMs can support both the equitable evaluation of teaching and teachers, and teacher training for pre-service and in-service educators. The evolution of GenAI is expected to make interaction with such systems increasingly intuitive [57].

At its core, the TEMPLES framework is designed as a multi-agent system that processes various educational artefacts to produce a robust, verified assessment. The process, illustrated in Figure 1, begins with the input of digital materials related to a course or teaching practice. While we envision that a future version of TEMPLEs might include passive sensors to monitor teaching in real-time, similar to PoT, the current framework is designed to operate on traditional digital artefacts. These inputs can include a wide variety of materials, such as lesson plans, lecture slides, transcripts of class activities, assignment descriptions, student engagement statistics from a learning management system (LMS), and so on.

Once the artefacts are submitted, the user provides an instruction, which can be for summative evaluation (e.g., "Evaluate the professional quality and appropriateness of the submitted artefacts with respect to the specified context and learning outcomes") or for formative development (e.g., "Provide constructive feedback on the input artefacts").

The instruction and artefacts are then passed to a panel of LLM sub-agents. Each sub-agent is configured with a distinct "personality" (e.g., 'critical pedagogue', 'assessment design expert', 'student engagement specialist') and provides its own critique and feedback.

The feedback from all sub-agents is then collated by a Master Agent that synthesises the various points into a single, coherent draft evaluation. However, this draft is not immediately presented to the user. Instead, it is passed to a specialised MT Verification Agent. This agent checks the Master Agent's synthesised suggestions for correctness and logical consistency by applying a set of pre-defined MRs. If the draft fails verification, it is returned to the Master Agent with a request for refinement. This request is specific and actionable. For example, it might identify a logical contradiction across the submitted materials, such as: "The synthesised feedback praises the lesson plan's focus on critical thinking, but the evaluation of the corresponding exam paper suggests it only tests for factual recall: Please reconcile this inconsistency." Alternatively, it could flag a violation of an expected relationship, stating: "A follow-up test case with simplified terminology was applied to the source lecture slides, yet the synthesised feedback did not register an improvement in the clarity dimension: Please re-evaluate this aspect."

The first example highlights a critical function of the MT Verification Agent: Enforcing constructive alignment [58]. This pedagogical principle asserts that assessment tasks (e.g., the exam) must be directly aligned with the intended learning outcomes (e.g., critical thinking) and supported by the teaching and learning activities (e.g., the lesson plan). A misalignment, where the assessment does not measure the stated goal, is a significant flaw in course design. By automatically detecting such inconsistencies, the TEMPLES framework provides feedback that is not just evaluative but also formative, helping educators to improve the fundamental design of their courses. This creates an iterative loop, where the Master Agent revises the feedback based on the MT agent's checks until an agreement is reached and the MRs are satisfied.

Finally, once the feedback has been successfully verified, the system presents a single, consolidated output to the user. Therefore, the final evaluation is not just a collection of opin-

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ions, but a robust assessment that has undergone a rigorous (but automated) verification process.

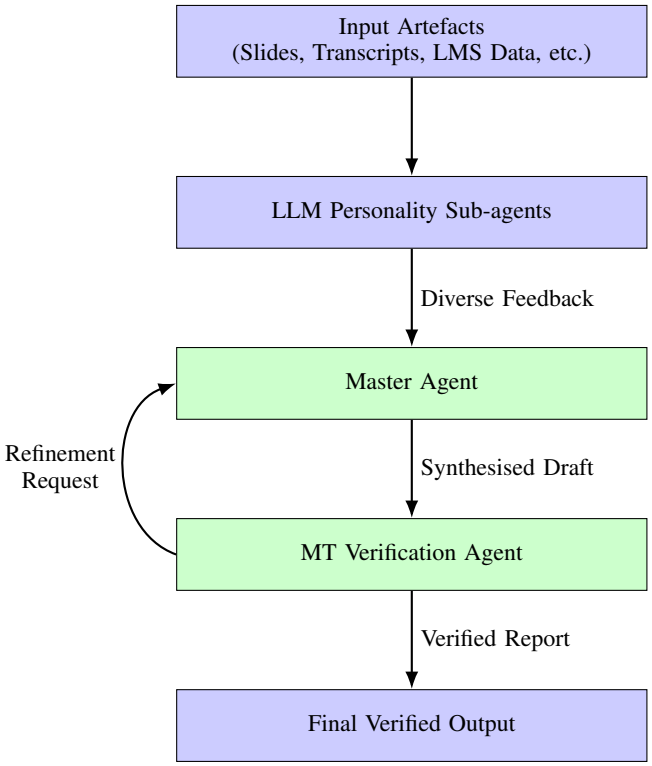


Fig. 1. The TEMPLES Framework Process

V. Conclusion

This paper reports on a work-in-progress: The development, and eventual implementation and deployment, of a framework and tool to support scalable, equitable evaluation of teaching (including teachers). The tool is anticipated to be shared and licensed as an OER (as CC BY [59], [60]), and made available through Merlot [61], and other repositories. In addition to supporting evaluation, it is hoped to also become a tool to support teacher training, both for pre-service teachers, and for the CPD of those already in service. As an initial framework, it is hoped that it will be of use to the community, and that the community will expand its scope and applicability. Our future work will include the framework's refinement and implementation, and also further exploration of how MT, and especially metamorphic thinking, can support teachers and students in the education experience.

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